

8-Week Web Development Course

Secondary School Coding Curriculum

This course is designed for secondary school students learning coding for the first time. The curriculum focuses on practical web development skills using HTML, CSS, and JavaScript. Students will learn how websites are created and build projects throughout the program.

Week	Topic
1	Introduction to Coding
2	HTML Basics
3	HTML Forms
4	CSS Basics
5	CSS Layout
6	JavaScript Basics
7	Mini Project
8	Final Project

Week 1 – Introduction to Coding

Coding means giving instructions to a computer to perform tasks. Programmers create websites, applications, games, and software systems. Students should understand: What coding is How websites work The difference between frontend and backend development Why coding is important in modern careers Important Terms: **Website:** A collection of web pages **Browser:** Software used to open websites **Frontend:** The visible part of a website **Backend:** The server-side part of a website Practical Activity: Students should open a browser and inspect different websites to identify menus, buttons, text, and images.

Week 2 – HTML Basics

HTML stands for HyperText Markup Language. It is the structure of every website. Students will learn: HTML document structure Headings Paragraphs Images Links Lists Basic HTML Example: `<!DOCTYPE html>`

```
<html>
```

```
<head>
```

```
<title>My First Website</title>
```

```
</head>
```

```
<body>
```

```
<h1>Welcome</h1>
```

```
<p>This is my first webpage.</p>
```

```
</body>
```

```
</html>
```

 Practical Exercise: Create a personal webpage containing: Your name Your favourite food Your hobbies

An image

Week 3 – HTML Forms

HTML forms are used to collect information from users. Students will learn: Input fields Text areas Buttons Checkboxes Radio buttons Example Form: `<form>`

Name: `<input type="text">`

```
<br/>
```

Email: `<input type="email">`

<button>Submit</button>

</form> Practical Exercise: Build a student registration form.

Week 4 – CSS Basics

CSS stands for Cascading Style Sheets. CSS is used to style and design websites. Students will learn: Colours Fonts Backgrounds Borders Spacing Example CSS: body {

background-color: lightblue;

font-family: Arial;

}

h1 {

color: darkblue;

} Practical Exercise: Style the webpage created in Week 2.

Week 5 – CSS Layout

CSS layout controls how items are arranged on a webpage. Students will learn: Flexbox Display property Margins and padding Navigation bars Simple responsive design Example Flexbox: .container {

display: flex;

justify-content: space-between;

} Practical Exercise: Create a webpage with: A navigation menu Three content boxes A footer section

Week 6 – JavaScript Basics

JavaScript makes websites interactive. Students will learn: Variables Functions Buttons and events Alerts Simple calculations Example JavaScript: <script>

function greet() {

alert("Welcome to Coding Class");

}

</script> Practical Exercise: Create a button that displays a welcome message when clicked.

Week 7 – Mini Project

Students should combine HTML, CSS, and JavaScript to build a small project. Suggested Mini Projects: Personal portfolio website Simple calculator Quiz webpage School information webpage Project Requirements: Use HTML structure Add CSS styling Add at least one JavaScript feature

Week 8 – Final Project

Students should work individually or in groups to create a complete website. Possible Final Projects: School website Restaurant website Music artist website Online store homepage Final Project Requirements: Multiple webpages Navigation menu Styled layout Interactive JavaScript feature Good design and organisation Presentation: Students should explain: What the website does Tools used Challenges faced What they learned

Teaching Tips for the Instructor

- Encourage students to practise daily.
- Use practical demonstrations during lessons.
- Allow students to ask questions frequently.
- Focus more on building projects than memorising notes.
- Celebrate students' creativity and progress.
- Use simple real-life examples to explain coding concepts.

Student Assessment Guide

Students can be assessed using: Class participation Assignments Practical coding exercises Mini project Final project presentation

Conclusion

By the end of this 8-week course, students should understand the basics of web development and be able to build simple websites using HTML, CSS, and JavaScript.